

2 Methods

The shellgame framework formalizes and flags the epistemic failure mode of treating geographic transformation as a sequence of decisions that alter the underlying data-generating process.

Two decision points are typically left implicit in applied work.

Decision 1: Membership Definition. Analysts must choose whether to define geographic membership by administrative linkage (relationship-based, as used in Census tabulations) or by geometric contact (spatial intersection). In the example geography used, Hennepin County, Minnesota, this choice affects whether 74 ZCTAs (relationship-based) or 94 ZCTAs (geometric intersection) constitute the baseline—before any transformation occurs. The relationship file reflects Census tabulation logic, not raw polygon contact. Geometric intersection, by contrast, performs a literal spatial test: if a ZCTA polygon touches the county boundary, it counts, including edge-touching, slivers, water boundaries, and TIGER topology artifacts. This decision alone altered the baseline by 27% before any crosswalk is applied. For this analysis, membership is defined using the relationship-based administrative linkage approach.

Decision 2: Crosswalk Selection. When allocating ZIP-level data to counties, analysts must select among allocation ratios. The HUD ZIP-County crosswalk provides TOT_RATIO (total addresses), RES_RATIO (residential addresses only), BUS_RATIO (business addresses), and OTH_RATIO (other addresses). Each ratio embodies different assumptions about how phenomena distribute across addresses that may not hold uniformly across urban, suburban, and rural contexts ([Michels et al., 2025](#)). The analyst who selects TOT_RATIO assumes that total addresses are proxy for population distribution; the analyst who selects RES_RATIO assumes residential addresses are more appropriate. These are not technical details—they are methodological choices that affect results.

2.2 Quantifying Perturbation

The framework quantifies $\Delta X(\text{VAR})$, defined as the change in an analytic variable induced solely by geographic transformation and allocation choices, holding the underlying data source constant:

$$\Delta X(\text{VAR}) = \text{VAR}_1 - \text{VAR}_0$$

Where VAR_0 is the baseline value at geography A, and VAR_1 is the post-transformation value. $\Delta X(\text{VAR})$ does not imply which representation is "correct," nor that zero delta is inherently desirable. It states only that the variable is not invariant under this transformation. A ΔX of zero would indicate that the variable survived transformation unchanged; a non-zero ΔX indicates perturbation. The magnitude and direction of that perturbation are empirical questions, not theoretical ideals.

2.3 Implementation via geoDeltaAudit

The geoDeltaAudit package (v0.1.0) implements the shellgame framework in R ($\geq 4.0.0$), depending on the tidyverse ecosystem (dplyr, tidyr, stringr) and tidycensus for Census data access. The transformation pipeline consists of sequential "hops," with perturbation recorded at each step.

Hop 1: ZCTA $\rightarrow$ ZIP. Baseline ZCTA-level data are joined to a ZIP-ZCTA association table.

Membership is defined using the relationship-based administrative linkage approach implemented through the ZIP–ZCTA association table described below.

Where a single ZCTA maps to multiple ZIP codes, the ZCTA value is allocated equally across associated ZIPs using equal-share allocation. This stage captures pre-allocation expansion: the increase in spatial units representing the geography prior to any weighting..

Hop 2: ZIP $\rightarrow$ County. ZIP-level allocated values are joined to the HUD ZIP-County crosswalk and weighted by TOT_RATIO (or alternative ratio). Values are summed to the county level, with optional filtering to a target county. The TOT_RATIO represents the proportion of addresses in a ZIP code that fall within a given county, based on USPS address data. This second hop introduces additional perturbation as values are redistributed across county boundaries. At this stage, the analyst is no longer working with the original statistical units; the analytical surface has already shifted, and subsequent measurements represent transformed proxies rather than the population of interest.”

The pipeline returns an object of class "shellgame_audit" containing: baseline and recovered totals; absolute and percentage perturbation; unit counts at each stage; pre-allocation expansion percentage; and a data frame of values redistributed to neighboring counties. Geographic crosswalks are treated as directional allocations rather than invertible transformations, ensuring that each hop is audited as a one-way operation with explicit reporting of loss and fan-out at each stage. This directional treatment is essential: a ZCTA-to-ZIP crosswalk cannot be reversed to create a valid ZIP-to-ZCTA crosswalk, and the audit respects this asymmetry.

2.4 Demonstration Case

Using Hennepin County, Minnesota (FIPS 27053) as the example. Baseline data were retrieved from the 2022 ACS 5-year estimates (variable B01001_001) at the ZCTA level ($n=74$) via tidycensus. The ZIP-ZCTA association table was obtained from the UDS Mapper crosswalk, and the HUD ZIP-County crosswalk was obtained from HUD User (Q4 2024). This case was selected to demonstrate quantifying transformation-induced perturbation in analytic variables within a large metropolitan context where boundary complexity is maximized.